1. Practice CRUD operations on Mongo.

1. Create two collections -- emp and dept.

```
test> use empdept
switched to db empdept
empdept> db.createCollection("emp");
{ ok: 1 }
empdept> db.createCollection("dept");
{ ok: 1 }
empdept>
```

2. Insert the emp and dept records.

```
empdept> db.dept.insertMany([
...   { deptno: 10, dname: "ACCOUNTING", loc: "NEW YORK" },
...   { deptno: 20, dname: "RESEARCH", loc: "DALLAS" },
...   { deptno: 30, dname: "SALES", loc: "CHICAGO" },
...   { deptno: 40, dname: "OPERATIONS", loc: "BOSTON" }
... ]);
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('69bbd718576d257c314eeb86'),
    '1': ObjectId('69bbd718576d257c314eeb87'),
    '2': ObjectId('69bbd718576d257c314eeb88'),
    '3': ObjectId('69bbd718576d257c314eeb89')
  }
}
empdept>
```

```
empdept> db.emp.insertMany([
...   { empno: 7369, ename: "SMITH", job: "CLERK", sal: 800, deptno: 20 },
...   { empno: 7499, ename: "ALLEN", job: "SALESMAN", sal: 1600, deptno: 30 },
...   { empno: 7521, ename: "WARD", job: "SALESMAN", sal: 1250, deptno: 30 },
...   { empno: 7566, ename: "JONES", job: "MANAGER", sal: 2975, deptno: 20 },
...   { empno: 7698, ename: "BLAKE", job: "MANAGER", sal: 2850, deptno: 30 },
...   { empno: 7782, ename: "CLARK", job: "MANAGER", sal: 2450, deptno: 10 },
...   { empno: 7839, ename: "KING", job: "PRESIDENT", sal: 5000, deptno: 10 },
...   { empno: 7844, ename: "TURNER", job: "SALESMAN", sal: 1500, deptno: 30 },
...   { empno: 7876, ename: "ADAMS", job: "CLERK", sal: 1100, deptno: 20 },
...   { empno: 7900, ename: "JAMES", job: "CLERK", sal: 950, deptno: 30 },
...   { empno: 7902, ename: "FORD", job: "ANALYST", sal: 3000, deptno: 20 },
...   { empno: 7934, ename: "MILLER", job: "CLERK", sal: 1300, deptno: 10 }
... ]);
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('69bbd73a576d257c314eeb8a'),
    '1': ObjectId('69bbd73a576d257c314eeb8b'),
    '2': ObjectId('69bbd73a576d257c314eeb8c'),
    '3': ObjectId('69bbd73a576d257c314eeb8d'),
    '4': ObjectId('69bbd73a576d257c314eeb8e'),
    '5': ObjectId('69bbd73a576d257c314eeb8f'),
    '6': ObjectId('69bbd73a576d257c314eeb90'),
    '7': ObjectId('69bbd73a576d257c314eeb91'),
    '8': ObjectId('69bbd73a576d257c314eeb92'),
    '9': ObjectId('69bbd73a576d257c314eeb93'),
    '10': ObjectId('69bbd73a576d257c314eeb94'),
    '11': ObjectId('69bbd73a576d257c314eeb95')
  }
}
empdept>
```

3. Increase salary of dept 10 emps by 100.

```
empdept> db.emp.updateMany(
... {deptno:10},
... {$inc:{sal:100}}
... );
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 4,
  modifiedCount: 4,
  upsertedCount: 0
}
empdept>
```

4. Display emp name, job, deptno, sal in sorted order of deptno, job, sal (desc) for all emps whose sal more than 1000.

```
empdept> db.emp.find(
...   { sal: { $gt: 1000 } },
...   { _id: 0, ename: 1, job: 1, deptno: 1, sal: 1 }
... ).sort({ deptno: 1, job: 1, sal: -1 });
[
  { ename: 'MILLER', job: 'CLERK', sal: 1400, deptno: 10 },
  { ename: 'CLARK', job: 'MANAGER', sal: 2550, deptno: 10 },
  { ename: 'KING', job: 'PRESIDENT', sal: 5100, deptno: 10 },
  { ename: 'FORD', job: 'ANALYST', sal: 3000, deptno: 20 },
  { ename: 'ADAMS', job: 'CLERK', sal: 1100, deptno: 20 },
  { ename: 'JONES', job: 'MANAGER', sal: 2975, deptno: 20 },
  { job: 'salesman', deptno: 20, sal: 1500 },
  { job: 'Clerk', deptno: 30, sal: 1900 },
  { ename: 'BLAKE', job: 'MANAGER', sal: 2850, deptno: 30 },
  { ename: 'ALLEN', job: 'SALESMAN', sal: 1600, deptno: 30 },
  { ename: 'TURNER', job: 'SALESMAN', sal: 1500, deptno: 30 },
  { ename: 'WARD', job: 'SALESMAN', sal: 1250, deptno: 30 }
]
empdept>
```

5. Delete the dept with id 40.

```
empdept> db.dept.deleteOne({ deptno: 40 });
{ acknowledged: true, deletedCount: 1 }
empdept>
```

2. Learn grouping operations using Mongo aggregation pipeline

1. Display job wise total sal, max sal, min sal, and avg sal for all jobs other than PRESIDENT.

```
empdept> db.emp.aggregate([
...   { $match: { job: { $ne: "PRESIDENT" } } },
...   {
...     $group: {
...       _id: "$job",
...       total_sal: { $sum: "$sal" },
...       max_sal: { $max: "$sal" },
...       min_sal: { $min: "$sal" },
...       avg_sal: { $avg: "$sal" }
...     }
...   }
... ]);
[
  {
    _id: 'salesman',
    total_sal: 1500,
    max_sal: 1500,
    min_sal: 1500,
    avg_sal: 1500
  },
  {
    _id: 'Plumber',
    total_sal: 250,
    max_sal: 250,
    min_sal: 250,
    avg_sal: 250
  },
  {
    _id: 'CLERK',
    total_sal: 4250,
    max_sal: 1400,
    min_sal: 800,
    avg_sal: 1062.5
  },
  {
    _id: 'SALESMAN',
    total_sal: 4350,
    max_sal: 1600,
    min_sal: 1250,
    avg_sal: 1450
  },
  {
    _id: 'Clerk',
    total_sal: 1900,
    max_sal: 1900,
    min_sal: 1900,
    avg_sal: 1900
  },
  {
    _id: 'MANAGER',
    total_sal: 8375,
    max_sal: 2975,
    min_sal: 2550,
    avg_sal: 2791.6666666666665
  },
  {
    _id: 'ANALYST',
    total_sal: 3000,
    max_sal: 3000,
    min_sal: 3000,
    avg_sal: 3000
  }
]
```

2. Display emp name, dept name, income (sal + comm) in sorted order of their income.

```
empdept> db.emp.aggregate([
...     {
...       $lookup: {
...         from: "dept",
...         localField: "deptno",
...         foreignField: "deptno",
...         as: "dept_details"
...       }
...     },
...     { $unwind: "$dept_details" },
...
...     {
...       $addFields: {
...         income: { $add: ["$sal", { $ifNull: ["$comm", 0] }] }
...       }
...     },
...
...     {
...       $project: {
...         _id: 0,
...         ename: 1,
...         dept_name: "$dept_details.dname",
...         income: 1
...       }
...     },
...
...     { $sort: { income: 1 } }
... ]);
[
  { income: 250, dept_name: 'ACCOUNTING' },
  { ename: 'SMITH', income: 800, dept_name: 'RESEARCH' },
  { ename: 'JAMES', income: 950, dept_name: 'SALES' },
  { ename: 'ADAMS', income: 1100, dept_name: 'RESEARCH' },
  { ename: 'WARD', income: 1250, dept_name: 'SALES' },
  { ename: 'MILLER', income: 1400, dept_name: 'ACCOUNTING' },
  { ename: 'TURNER', income: 1500, dept_name: 'SALES' },
  { ename: 'ALLEN', income: 1600, dept_name: 'SALES' },
  { income: 1700, dept_name: 'RESEARCH' },
  { income: 1900, dept_name: 'SALES' },
  { ename: 'CLARK', income: 2550, dept_name: 'ACCOUNTING' },
  { ename: 'BLAKE', income: 2850, dept_name: 'SALES' },
  { ename: 'JONES', income: 2975, dept_name: 'RESEARCH' },
  { ename: 'FORD', income: 3000, dept_name: 'RESEARCH' },
  { ename: 'KING', income: 5100, dept_name: 'ACCOUNTING' }
]
empdept>
```